

MATTEO CIRILLO

MSc Student in Microengineering & Quantum Science

+41 79 685 34 73
matteo.c@bluewin.ch
[LinkedIn](#)
Switzerland



SUMMARY

EPFL Master's student, specializing in nanotechnology, with over a year of microfabrication experience in cleanroom environments. Held two research assistant roles working on all round design, fabrication, and characterization of MEMS devices for Scanning Probe Microscopy and Quantum Computing. Further completed various internships at ETH Zurich, Single Quantum, and NASA JPL, all with a primary focus on superconducting devices and photonics.

EDUCATION

Master of Science	EPFL - Swiss Federal Institute of Technology Major in Microengineering and Minor in Quantum Science (GPA : 97%)	2023 - 2026
Academic Exchange	TU Delft - Delft University of Technology Faculty of Electrical Engineering, Mathematics and Computer Science (GPA : 90%)	2021 - 2022
Bachelor of Science	EPFL - Swiss Federal Institute of Technology Major in Microengineering (GPA : 88%)	2019 - 2022

WORK EXPERIENCE

Research Fellow,	NASA - Jet Propulsion Laboratory / California Institute of Technology - Master's thesis on the design and testing of Superconducting Nanowire Single Photon Detectors (SNSPDs), for applications in Deep Space Optical Communications and Photonic Quantum Computing. - Performed systematic studies of photoresponses to improve both the fabrication processes of the superconducting thin films (WSi, NbTiN, MgB ₂), and the theoretical understanding of the devices' physics.	Jan 2026 – Sep 2026
Intern in R&D,	Single Quantum B.V. - Led a theoretical and experimental investigation of blackbody radiation induced dark counts in Superconducting Nanowire Single Photon Detectors (SNSPDs) Systems. - Developed a physical model and simulation framework to quantify thermal photon generation and propagation through the SNSPD systems. This now supports the design of low dark count rate products. - Designed and validated strategies to mitigate dark counts at telecom wavelength, thus improving suitability for applications in Quantum Key Distribution (QKD) and Optical Communications.	Oct 2025 – Dec 2025
Summer Intern,	ETH Zurich / PSI - Trapped Ion Quantum Information Group - Joined the Ion Trap Quantum Computing (ITQC) lab and designed a compact optical system for double-pass acousto-optic modulation. The device is now deployed in various setups at ETH Zurich and PSI. - Optimized the design for laser frequency scanning and switching, while achieving a 300% reduction in size compared to the optical table prototype. Also broadened wavelengths compatibility to 300 – 900nm.	Jul 2024 – Sep 2024

RELEVANT PROJECTS

Fabrication of Trilayer AFM Cantilevers for Scanning Electrochemical Microscopy Applications.	Feb 2025 – Jul 2025
- Semester thesis at EPFL's LBNI group, on the design and microfabrication of composite (SiN, BCB, Cr, Au, Ti, Pt) atomic force microscopy cantilevers for scanning electrochemical microscopy applications. - Fabricated the designed devices in cleanroom and experimentally demonstrated novel versatile scanning probe capabilities that resulted in a collaboration with leading industry partners.	
Processes for Niobium Resonators in Circuit Quantum Electrodynamics.	Jul 2024 – Feb 2025
- Semester thesis at EPFL's HQC lab, on the microfabrication and testing of superconducting microwave resonators in Niobium, with a focus on improving internal quality factors. - Identified critical flaws in the thin film processing, providing insights for improved device robustness. - Performed both cleanroom microfabrication and radio frequency characterization in a dilution cryostat.	

SKILLS, CERTIFICATIONS & AWARDS

Languages	English (<i>Fluent: C2</i>), French (<i>Native</i>), Italian (<i>Native</i>), German (<i>Intermediate: B2</i>)	
Technical	Cleanroom microfabrication (Etching, PVD, DUV lithography, Metrology), Optical design, CAD design (CATIA, Altium), FEM Simulation (COMSOL), Mechanical machining, Safety protocols (Laser and chemicals).	
Coding	Python (and data science packages), Matlab, Mathematica, LabView. Intermediate with C++.	
JPL Graduate Fellowship Awardee,	NASA JPL / California Institute of Technology	2026
C2 Certificate in Advanced English,	Cambridge University	2019
Prize for Excellence in Mathematics and Sciences,	Nestlé Switzerland S.A.	2016

OTHER ACTIVITIES

Teaching	Tutoring students up to graduate level. Working freelance, for professors, and private.
Hobbies	Volleyball player, endurance runner, guitarist, and mathematics hobbyist.